

Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of global trends in primary energy and electricity

Audience: Policy analysts and energy sector strategists

Focus: Progress, technology mix, and regional transition gaps

75% of Global GHG Emissions Still Come from Fossil Fuels

Fossil fuels have dominated energy systems since the Industrial Revolution.

Three-quarters of global greenhouse gas emissions come from fossil fuel combustion for energy.

Air pollution from fossil fuels causes at least 5 million premature deaths annually.

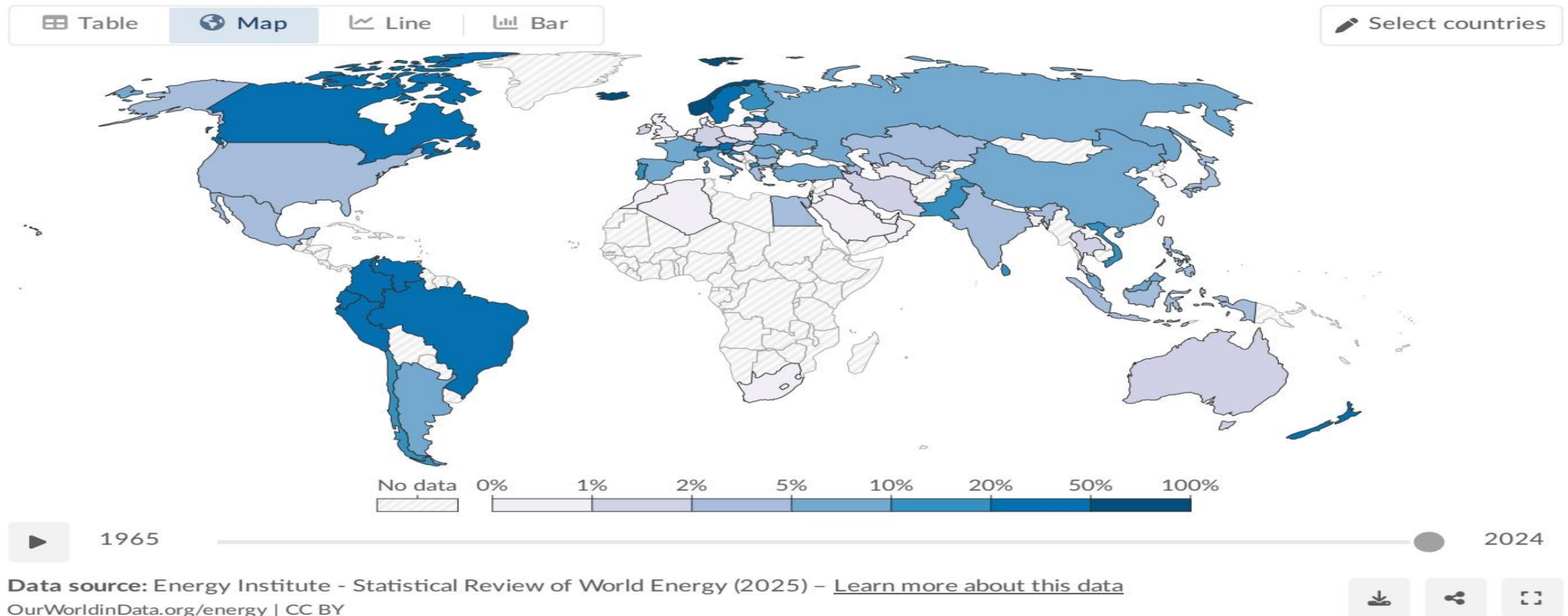
Rapid scale-up of low-carbon energy is essential for climate and health outcomes.

Urgency Signal

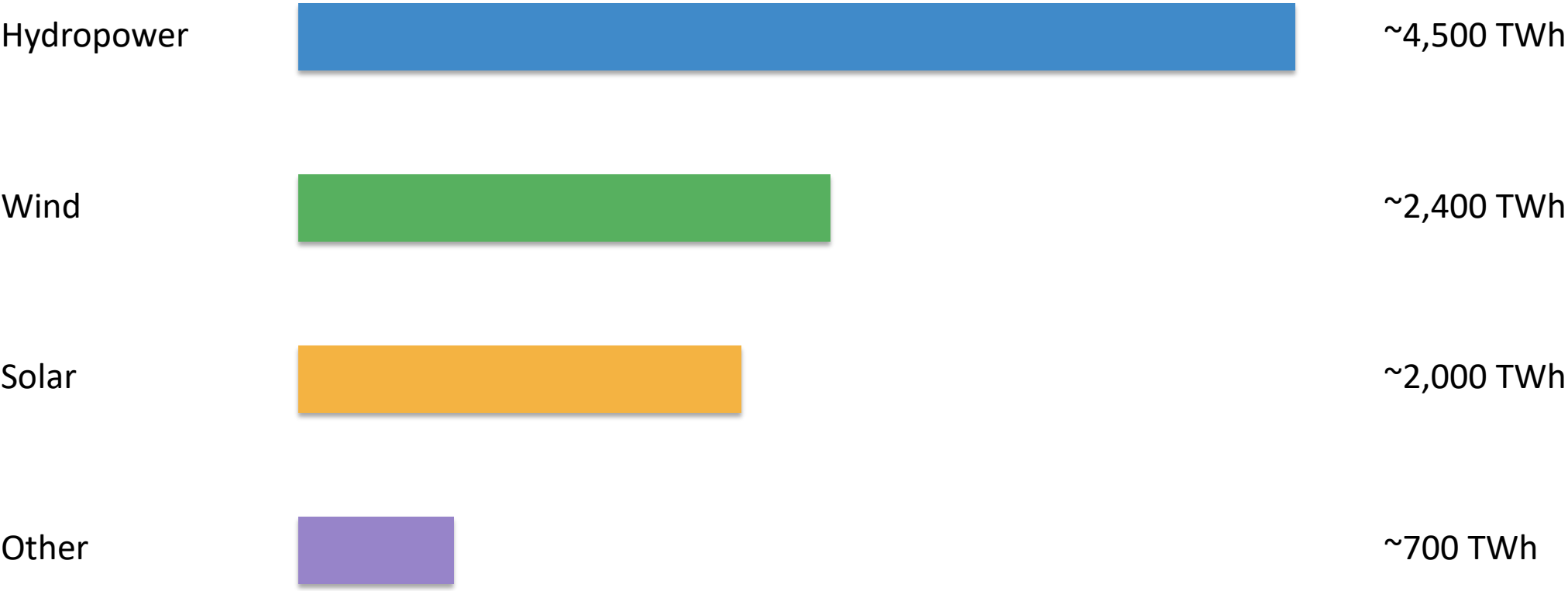
75% GHG from fossil fuels
5M+ premature deaths/year
Transition pace still insufficient

Renewables Now Supply ~14% of Global Primary Energy

2024 estimate (substitution method): roughly one-seventh of global primary energy is renewable.



Renewable Electricity Generation Is Growing Rapidly, Led by Win



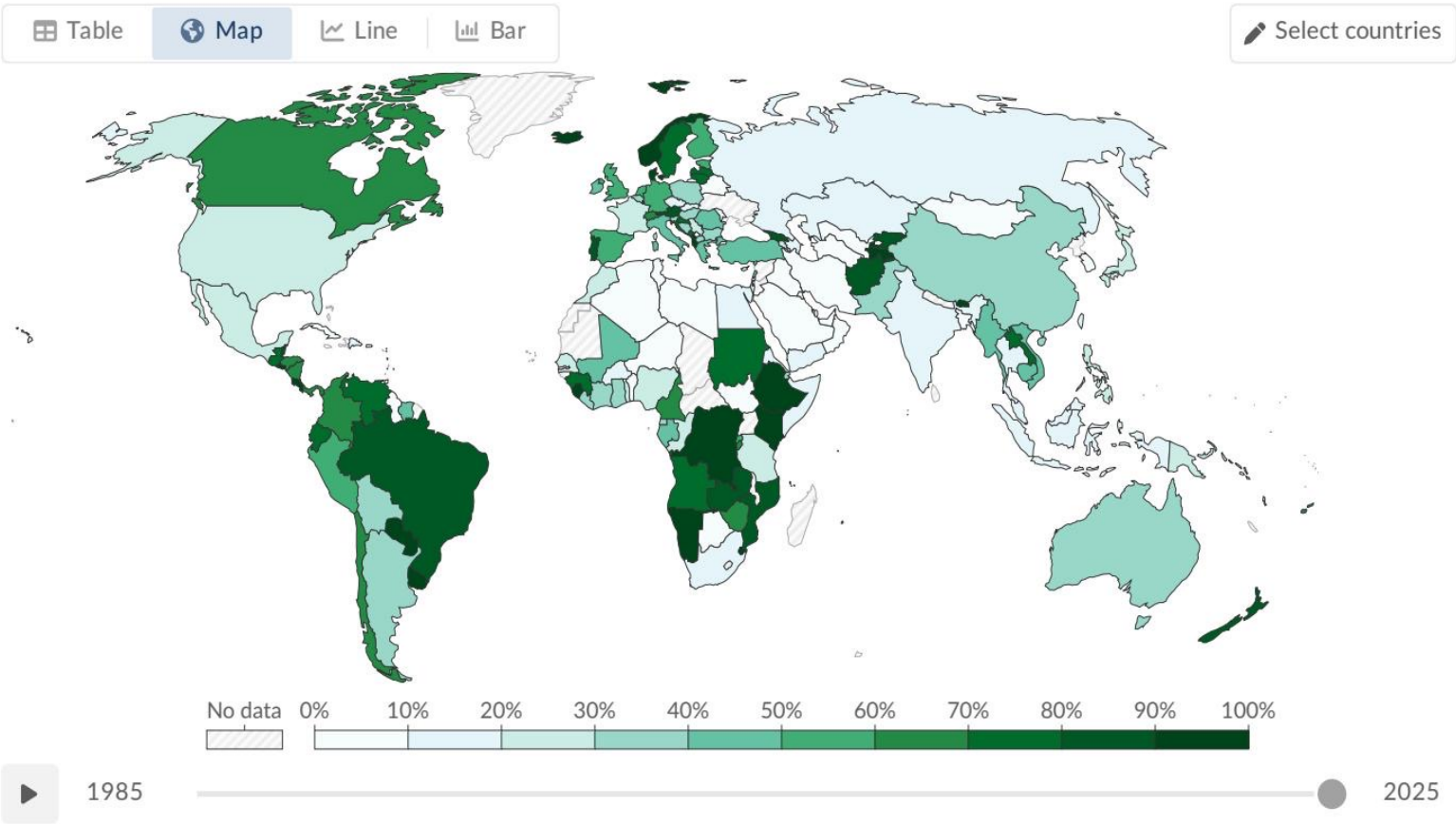
Wind + solar (~4,400 TWh) now nearly match hydropower (~4,500 TWh).

Source: OWID brief table (2024 generation by technology)

Almost One-Third of Global Electricity Now Comes from Renewal

Share of electricity production from renewables, 2025

Renewables include solar, wind, hydropower, bioenergy, geothermal, wave, and tidal sources.



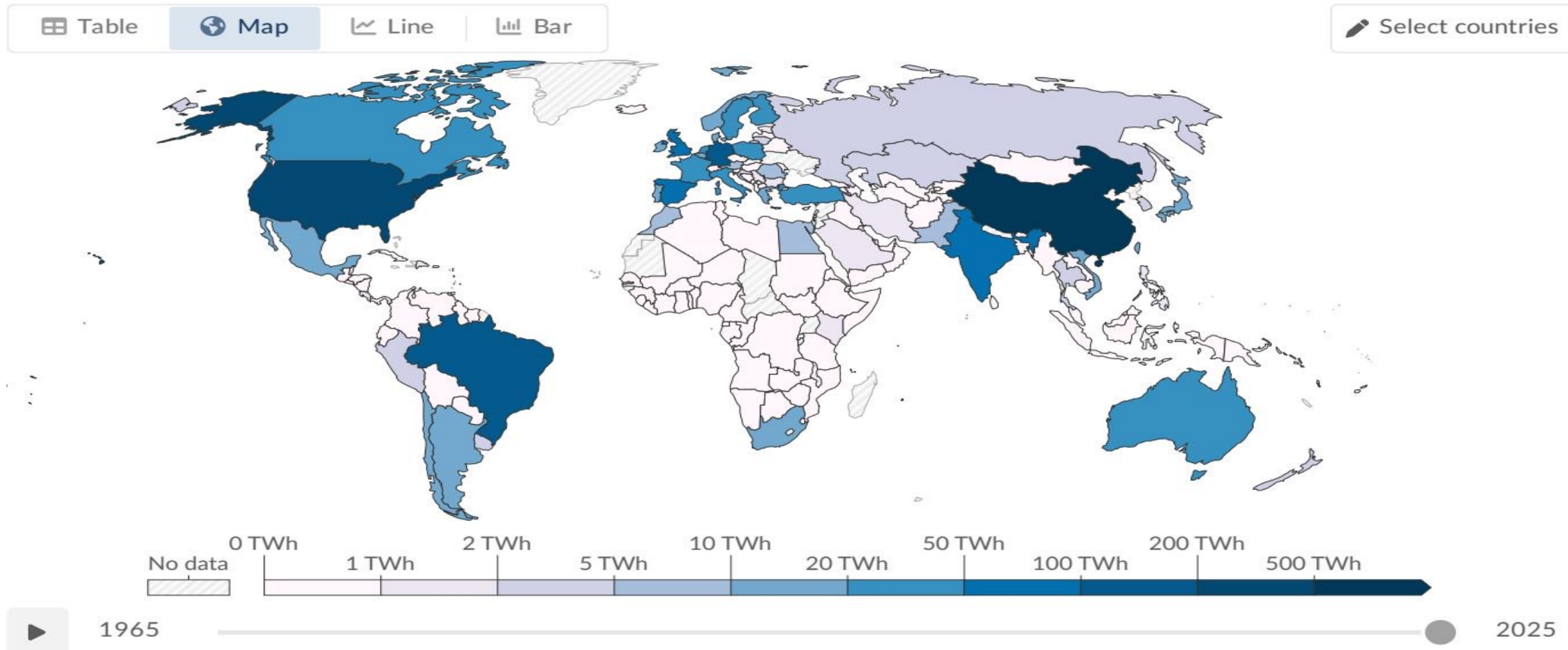
Data source: Ember (2026); Energy Institute - Statistical Review of World Energy (2025) – [Learn more about this data](#)
OurWorldinData.org/energy | CC BY



Country/Region	Renewable Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%

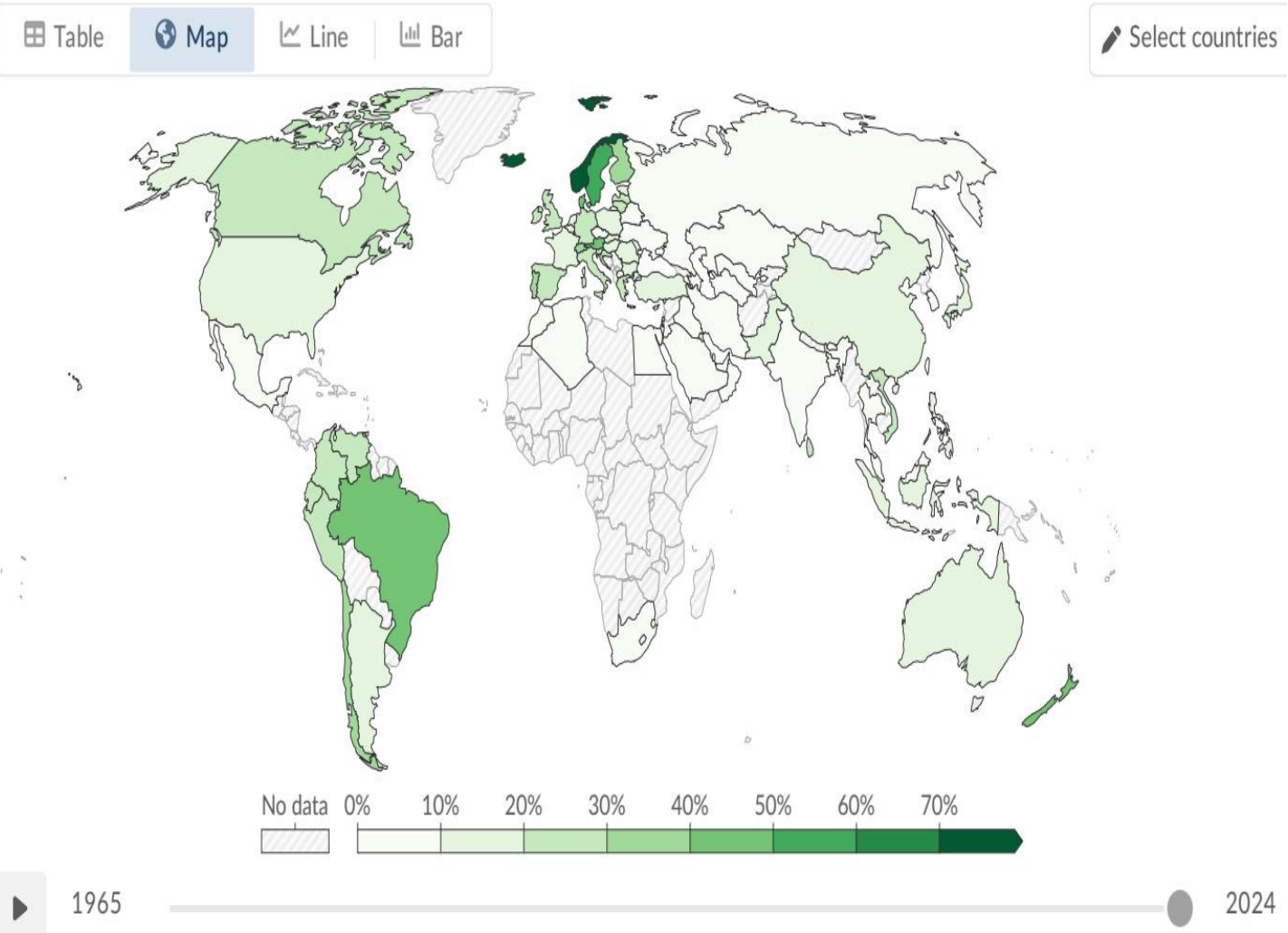
Hydropower Remains the Largest Renewable Source at ~50% of C

Top producers include China, Brazil, and Canada; hydro remains the backbone of renewable electricity.



Source: OWID figure_6 — Hydropower generation by country (2025)

Wind and Solar Are the Fastest-Growing Energy Technologies in H



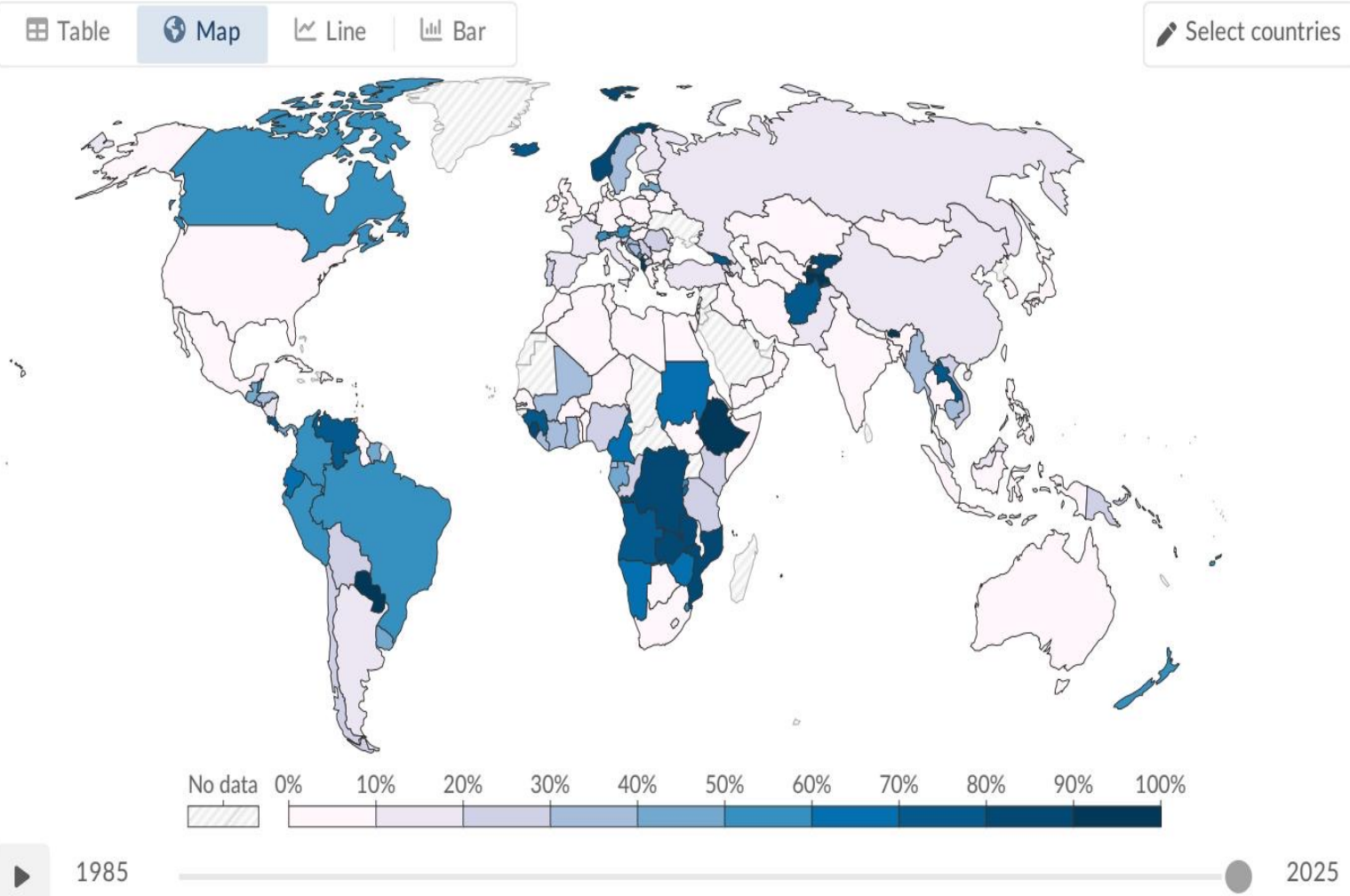
Technology	Generation (TWh)	Share of renewable electricity
Wind	~2,400	~25%
Solar	~2,000	~21%

Solar shows the steepest post-2010 growth trajectory.

Wind + solar together now nearly equal hydropower output.

Source: OWID figure_8 and technology shares from source brief

Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia a



Scandinavia, Brazil, and Canada often reach 65–98% renewable electricity.

Several African and Middle Eastern countries remain below 10%.

Hydropower endowments drive leadership in some regions.

Investment gaps and grid constraints slow transition in lower-income markets.

Key Takeaways: Renewables Are Growing Fast but the Transition

~14% of global primary energy and ~30% of global electricity now come from renewables.

Hydropower still dominates renewable generation, but wind and solar are catching up rapidly.

Wind and solar combined now nearly match hydropower output.

Regional disparities remain stark: leaders exceed 60–90%, while lagging regions remain below 10%.

Transport and heating still lag electricity, requiring faster policy and investment action.

Forward look: Accelerating grids, storage, and clean end-use electrification is now the strategic priority.